PROVGUARD

Confusion-selected benign data for prompt-injection detectors

01 THE FAILURE

High IID accuracy can hide benign over-defense

98.15% IID F1

20.67% External benign FPR

Observed benign stress example

Help me write a system prompt for a coding assistant that resists indirect prompt injection attempts.

Flagged by baseline: 5/5 seeds
Flagged by ProvGuard: 0/5 seeds

Released curated example; not an external-set exemplar.

02 TARGETED TRAINING DATA

Real candidate prompts
Dolly + OASST1; lexical leakage controls

↓

Baseline ensemble scoring
Mean injection probability; seeds 13 / 42 / 123

↓

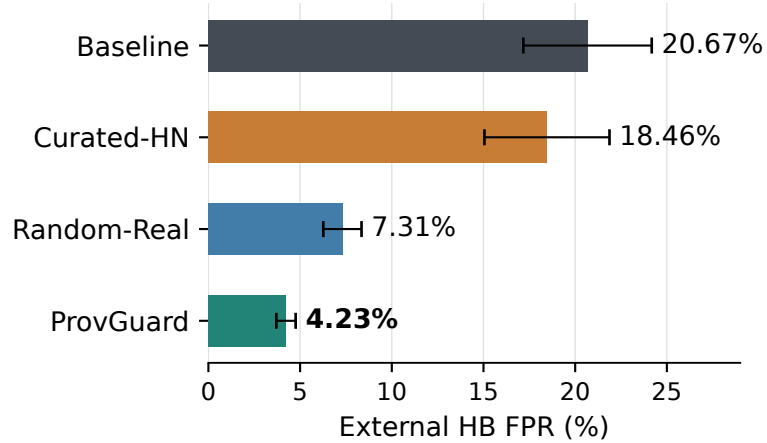
Rank + semantic validity
Two separate automated passes; both BENIGN

↓

Top-K + retrain same detector
K=1000 primary; no architecture change

Fixed threshold 0.5; five training seeds per arm.

03 EXTERNAL-SUBSET RESULT



Attack recall: 98.71% -> 98.69%

Random-Real external gain supported; mixed-set advantage remains inconclusive.

N=208 external prompts; bars show mean +/- sample SD.

Targeted real hard negatives reduce external over-defense while retaining attack detection.